

HIGH-LEVEL DESIGN (HLD) DOCUMENT

Project: CleanAir — Intelligent AC Vent Fan Controller

Target Hardware: STM32F446RE (Nucleo-64)

Operating System: FreeRTOS

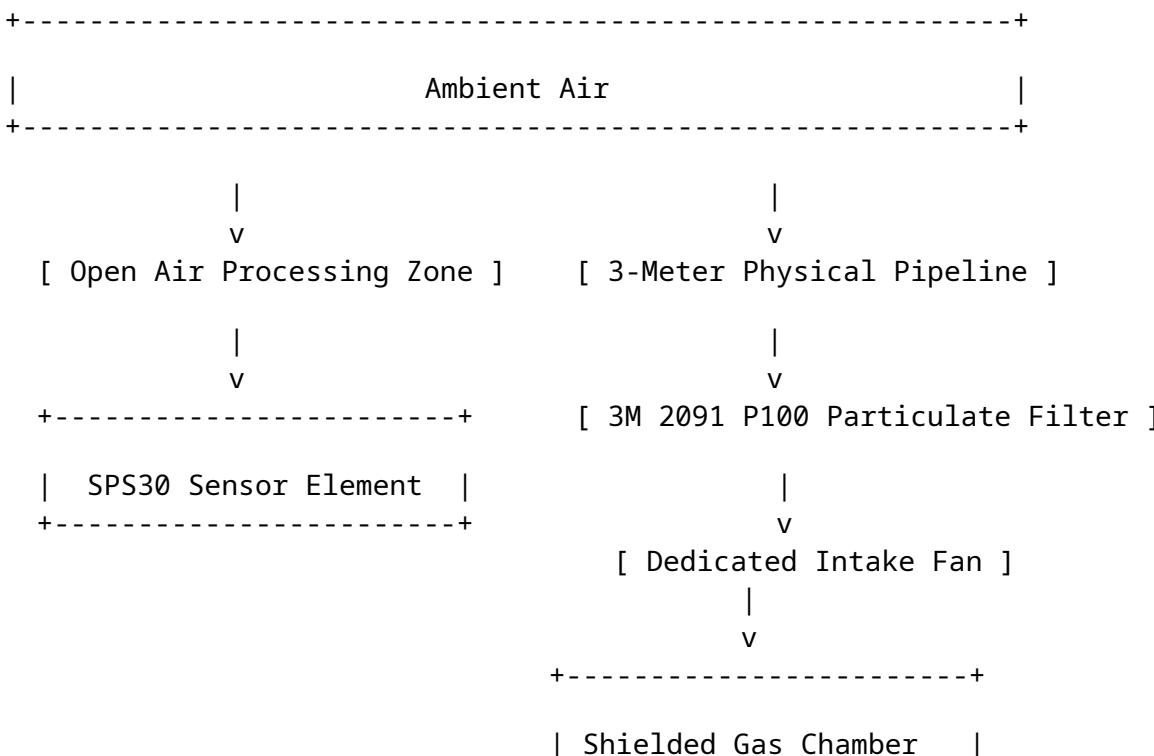
1. Document Overview

This High-Level Design (HLD) document describes the architecture, software layout, and operational logic of the CleanAir environmental control system. The system monitors ambient air particulates and shielded hazardous electrode gases, dynamically regulating an industrial AC ventilation fan using a solid-state TRIAC interface.

2. System Architecture & Physical Deployment

2.1 Environmental Zoning & Isolation Matrix

The system employs a dual-zone sampling architecture to isolate sensitive gas-sensing elements from destructive particulate contamination while maintaining accurate ambient monitoring.



	-	MQ131	-	MiCS-4514	
	-	BME680	(BSEC2 Logic)		
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- **Ambient Air Zone:** The SPS30 sensor sits in raw, unshielded air to immediately capture environmental particulate matter (PM2.5 and PM10) before any filtration occurs.
- **Shielded Gas Chamber:** The MQ131, MiCS-4514, and BME680 sensors are housed inside an isolated testing volume. Air is pulled through a 3-meter tube by an active intake fan and passed through a 3M 2091 P100 filter. This filter captures 99.97% of airborne dust and metallic electrode debris, letting only clean target gases pass over the metal-oxide elements.

2.2 System Power Architecture

- **Microcontroller & Sensors:** Powered by an external, regulated 5V power supply wired directly to the Nucleo's `E5V` input pin. The onboard power selection jumper (`JP5`) must be set to the `E5V` position.
- **Sensor Heating & Isolation:** The internal 5V heaters for the MQ131 and MiCS-4514 sensors run directly from the external 5V supply line to prevent electrical noise from disturbing the STM32's internal analog rails.
- **Logic Level Integrity:** The I²C2 bus uses 4.7 kΩ pull-up resistors tied exclusively to the Nucleo's 3.3V rail. Analog outputs from 5V sensor circuits pass through hardware voltage dividers to safely drop signals below the 3.3V maximum analog input threshold.

3. Target Gases & Environmental Indices

This system targets the specific physical and chemical byproducts generated during industrial welding, EDM (Electrical Discharge Machining), and plasma electrode processes.

3.1 Sensor Target Matrix & Application Coverage

- **SPS30 (Particulate Matter Sensor)**
 - *Target Media:* Mass concentrations of PM1.0, PM2.5, PM4.0, and PM10.
 - *Electrode Application:* Captures heavy metallic fumes, hexavalent chromium particulates, manganese dust, and airborne ash shedding from vaporized electrodes before they choke downstream filters. [1]
- **MQ131 (Ozone Gas Sensor)**
 - *Target Media:* Ozone (O_3) gas concentrations (low/high concentration variants).

- *Electrode Application:* Detects high-concentration O_3 produced by intense ultraviolet (UV) radiation from open electric arcs ionizing ambient oxygen molecules. [1, 2]
- **MiCS-4514 (Dual Metal-Oxide Sensor Channel)**
- *Target Media (CO Channel):* Carbon Monoxide (CO).
- *Target Media (NO_2 Channel):* Nitrogen Dioxide (NO_2).
- *Electrode Application:* Monitors toxic shielding gas breakdown products, incomplete combustion gases (CO), and high-temperature thermal fixation byproducts (NO_2) escaping the spark gap.
- **BME680 via Bosch BSEC2 (Environmental & Gas Engine)**
- *Target Media:* Volatile Organic Compounds ($VOCs$), Sulfur Compounds, Hydrogen Gas (H_2).
- *Electrode Application:* Tracks volatilized vapor complexes from cutting fluids, dielectric oils, electrode coatings, and insulating resins.

3.2 Implemented Environmental Indices

The control architecture processes individual gas sensor curves alongside BSEC2 data arrays to compute continuous safety indexes:

- **Chamber Air Quality Index (AQI):** A multi-variable index fusing CO , NO_2 , and BSEC2 VOC tracking into a singular danger metric to adjust ventilation baseline speeds.
- **Oxidizing Gas Index (OGI):** A high-priority control index combining MQ131 (O_3) and MiCS-4514 (NO_2) streams, reflecting extreme electrode ionization states.
- **Filter Loading Index (FLI):** Derived from the differential ratio between open-air SPS30 PM readings and chambered BME680 barometric pressure drop trends, signaling when the 3M 2091 filter is physically packed with debris.

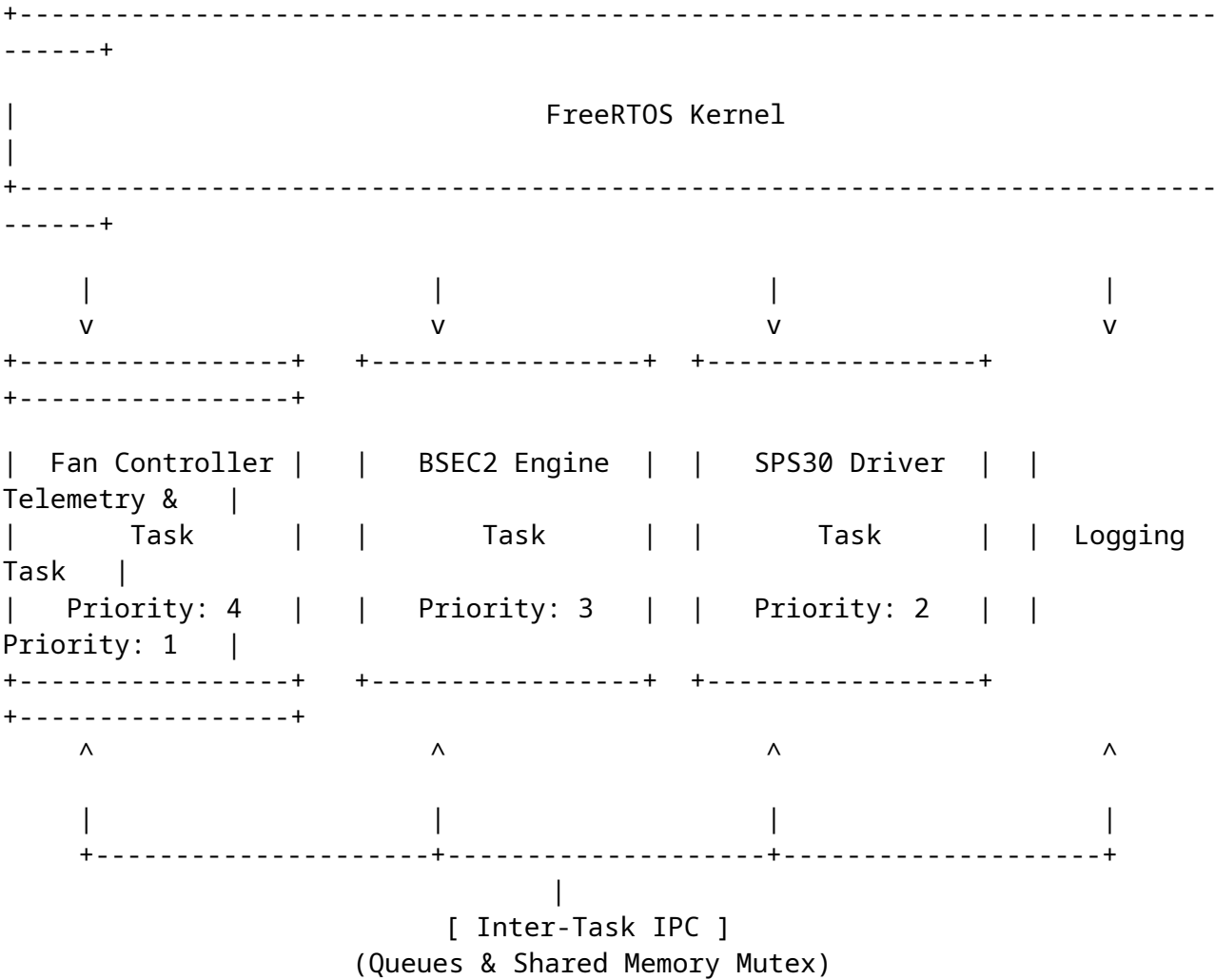
4. Hardware Peripheral Mapping

Subsystem Component	Peripheral Identifier	Physical Hardware Pin	Hardware Mode & Execution Profile
SPS30 + BME680	I2C2	PB10 (SCL), PB3 (SDA)	Standard Open-Drain. JTAG-SWO trace disabled on PB3.
MQ-131 Output	ADC1_IN0	PA0	Single-Ended Analog Input (Requires External Divider).
MiCS-4514 (CO)	ADC1_IN1	PA1	Single-Ended Analog Input (V_{OUT1}).
MiCS-4514 (NO_2)	ADC1_IN7	PA7	Single-Ended Analog Input (V_{OUT2}).

RobotDyn ZC Input	EXTI16	Pc6	Digital Input, configured for falling-edge interrupts.
RobotDyn Gate Out	GPIO Output	pc7	Push-Pull, High-Speed Output driven by Timer ISR.
Phase-Delay Tracking	TIM3	Internal	One-Pulse Hardware Mode (Triggered by ZC Interrupt).
OS Kernel Clock	SysTick	Internal	Dedicated exclusively to FreeRTOS Scheduler operations.
HAL Timebase	TIM6	Internal	Dedicated strictly to standard HAL delay and timeout loops.
Debug & Telemetry	USART2	PA2 (TX), PA3 (RX)	Asynchronous communication mapped to ST-LINK VCP.
Status Indicator	GPIO Output	PA5	Mapped to Nucleo User LED (LD2).

5. Software Architecture & Task Concurrency

The firmware is structured around a priority-preemptive FreeRTOS kernel. Dedicated, isolated software tasks balance long sensor communication windows with fast, real-time AC phase-angle tracking.



5.1 FreeRTOS Task Matrix

1. **Fan Controller Task (Priority 4 - High):** Computes required ventilation intensity using multi-sensor inputs and writes new delay values to the phase-control variables. Monitors system safety states.
2. **BSEC2 Engine Task (Priority 3 - Medium-High):** Runs the Bosch BSEC2 proprietary analysis library. Samples I2C2 at a fixed frequency to read the BME680, computing baseline-corrected temperature, humidity, pressure, and air quality metrics. Given its heavy math workload, it uses a larger stack size (4096 bytes).
3. **SPS30 Driver Task (Priority 2 - Medium):** Periodic worker thread that communicates with the SPS30 via I2C2. Pulls mass concentration metrics (PM2.5, PM10) every 1–2 seconds.
4. **Telemetry & Logging Task (Priority 1 - Low):** Formats collected sensor profiles, active phase delays, and error registers into clean text data streams. Outputs this information via USART2 to the ST-LINK Virtual COM Port.

5.2 Data Processing & Interrupt Flow

Analog data capture and AC phase timing use low-overhead hardware mechanisms to bypass OS scheduling delays:

- **Continuous Analog Capture:** ADC1 operates in Scan Mode via Direct Memory Access (DMA). It continuously updates a local 3-element RAM array with fresh raw voltages from the MQ131 and MiCS-4514 channels without shifting processing overhead onto the CPU.
 - **Zero-Cross & Phase Tracking Loop:**
 1. The AC line passes through a zero-voltage state, pulling the RobotDyn Zero-Cross pin (PA10) low.
 2. An external interrupt (EXTI10) triggers immediately, preempting the OS to signal the arrival of a new AC half-cycle.
 3. The EXTI handler initializes and starts TIM2 in One-Pulse Mode, loading the current phase delay value.
 4. When TIM2 matches the delay value, its hardware interrupt fires, driving PA6 high for a brief, fixed period to pulse the TRIAC gate and engage the fan.
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6. Mathematical & Physical Normalization

6.1 Pneumatic Lag & Transport Delay Compensation

Because the 3-meter intake pipe and P100 filter restrict flow, gas molecules reach the measurement chamber several seconds after the SPS30 encounters the ambient air. The processing engine matches ambient particulate readings with chambered gas concentrations by routing the open-air data through a software-managed ring buffer. This matches delayed chamber signals with the corresponding historical ambient data packet.

6.2 Environmental Gas Calibration

Changes in pressure and temperature inside the closed chamber alter the baseline resistance (R_0) of the metal-oxide gas sensors. The control system uses live temperature, relative humidity, and barometric pressure data calculated by the BSEC2 engine to dynamically scale the raw voltage curves (V_{OUT}) to PPM) for the MQ131 and MiCS-4514 channels.

7. Safety & Fail-Safe Operations

- **Loss of Signal Safety:** If the I2C2 bus locks up, or if the DMA stream drops out for more than 5 consecutive execution cycles, the system enters an Emergency Override state. It immediately sets the TRIAC phase delay to 0%, running the exhaust fan at 100% capacity to ensure continuous ventilation.
 - **Sensor Degradation Recovery:** If any gas sensor reads an impossible voltage (0V due to short circuit or 3.3V due to broken wire), the system flags a sensor failure error over USART2 and defaults the ventilation loop to maximum speed.
 - **BSEC2 State Saving:** To maintain sensor accuracy across reboots, the system backs up the BSEC2 algorithm's calibration matrix to the STM32's non-volatile internal flash memory once per hour.
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